Lecture 3 2025-09-04

Proposition: (from last time) Suppose $f \in C([a,b])$ and $g(x) = |f(x)|$. Then $g \in C([a,b])$ and the extrema of g occur at extrema or zeros of f.

Proof: Continuity: at x_0 ✓ **Want to show:**

$\forall \varepsilon > 0 \quad \exists \delta > 0$ s.t. $|x - x_0| < \delta$

then $|g(x) - g(x_0)| < \varepsilon$

Let $\varepsilon > 0$ arbitrary. Then since we know that f is continuous we can find $\delta > 0$ s.t. $|f(x) - f(x_0)| < \varepsilon$ whenever $|x - x_0| < \delta$.

Claim: Some δ works for g:

Let any ✓ x s.t. $|x - x_0| < \delta$

$$|g(x) - g(x_0)| = ||f(x)| - |f(x_0)|| \leq |f(x) - f(x_0)| < \varepsilon$$

($\leq$: Δ-inequality)

$\Rightarrow$ g is continuous

- If g has a maximum at c_2, then $g(x) \leq g(c_2)$ for $x \in [a,b]$

  That means $|f(x)| \leq |f(c_2)|$ for all $x \in [a,b]$

  and $-|f(c_2)| \leq f(x) \leq |f(c_2)|$

  If $f(c_2) \geq 0$ then $f(x) \leq f(c_2) = \max f$ for all $x \in [a,b]$

  $\Rightarrow$ c_2 is a maximum and therefore extrema of f

  If $f(c_2) < 0$ then $-|f(c_2)| = f(c_2) \leq f(x)$ for all $x \in [a,b]$

  $\Rightarrow$ c_2 is a minimum and hence extrema of f.

- Next let's consider minima of g.

Option 1: f has a zero $\iff \exists\ c \in [a,b]$ such that $f(c) = 0$

$\iff g(c) = 0 \leq g(x)$ for all $x \in [a,b]$

$\Rightarrow$ g has a minimum at a zero of f.

Option 2: f has no zeros. By IVT, f has constant sign on $[a,b]$

* If $f(x) > 0$ on $[a,b]$ then $g(x) = f(x)$ on $[a,b]$

and therefore f and g have the same extrema.

* If $f(x) < 0$ on $[a,b]$ then $g(x) = -f(x)$ on $[a,b]$

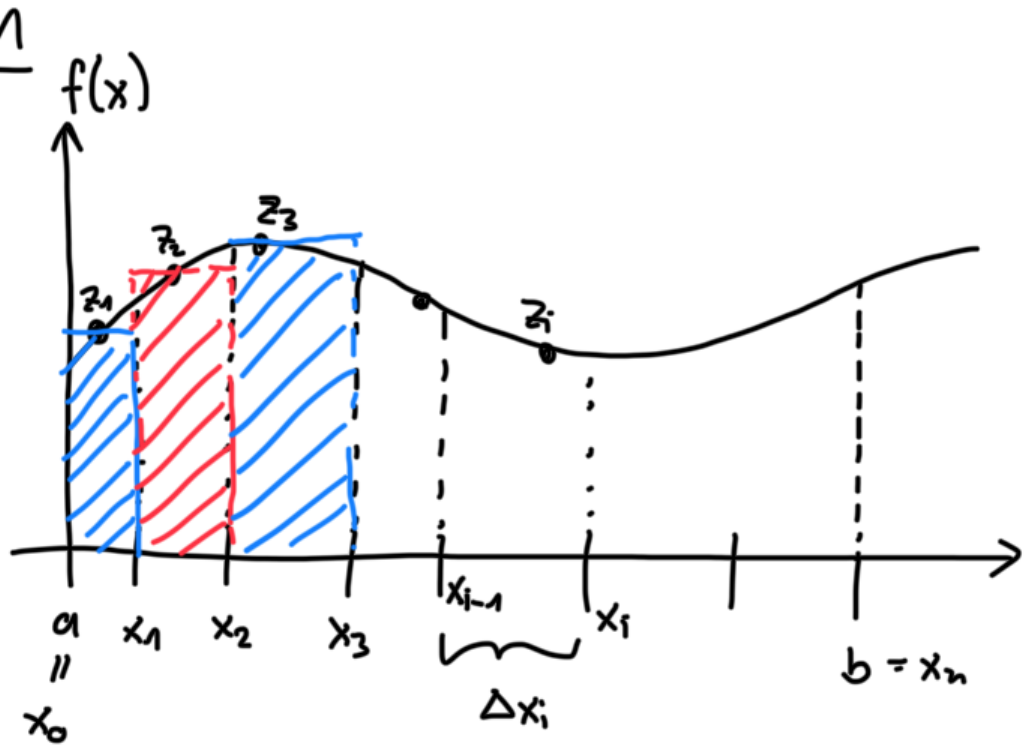
so if $f(c_1) \leq f(x) \leq f(c_2) \Rightarrow g(c_1) \geq g(x) \geq g(c_2)$

↑ multiply by -1

So g has a min (max) where f has a max (min). ∎

↑ end of proof q.e.d.

Integration



Def. 1.12 The Riemann integral of a function f on the interval $[a,b]$ is the following limit, provided it exists

$$\int_a^b f(x)\,dx = \lim_{\max\{\Delta x_i\} \to 0} \sum_{i=1}^{n} f(z_i)\,\Delta x_i$$

where $a = x_0 \le x_1 \le \dots \le x_n = b$ and $\Delta x_i = x_i - x_{i-1}$ for each $i = 1, 2, \dots, n$ and z_i are arbitrarily chosen in the interval $[x_{i-1}, x_i]$.

Thm 1.13 **Weighted Mean Value theorem for Integrals**

Suppose $f \in C([a,b])$, the Riemann integral of g exists on $[a,b]$ and that g does not change signs on $[a,b]$.

Then there exists a number $c \in (a,b)$ with

$$\int_a^b f(x)\,g(x)\,dx = f(c)\int_a^b g(x)\,dx .$$

Taylor's theorem / Thm 1.14 VERY IMPORTANT!

Suppose $f \in C^n([a,b])$ and $f^{(n+1)}$ exists on $[a,b]$.
Let $x_0 \in [a,b]$. Then for all $x \in [a,b]$ there exists $\xi(x)$ between x_0 and x such that

$$f(x) = P_n(x) + R_n(x)$$

where

$$P_n(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2} f''(x_0)(x-x_0)^2 + \dots + f^{(n)}(x_0)\frac{(x-x_0)^n}{n!}$$

$$= n\text{-th Taylor polynomial}$$

and

$$(*) \qquad R_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi(x))(x-x_0)^{n+1}$$

= remainder term / truncation error associated with $P_n(x)$.

Remarks:
- ξ depends on x but it may not be possible to determine it exactly.
- $(*)$ expresses R_n as the next term in the expansion but with $f^{(n)}(x_0)$ replaced by $f^{(n)}(\xi(x))$
- If $f \in C^\infty([a,b])$, the infinite series obtained by taking the limit $n \to \infty$ of $P_n(x)$ is called Taylor series of f around x_0. If $x_0 = 0$ it is called Maclaurin series.

- When $n=0$, Taylor's theorem is the MVT

$$f(x) = \underbrace{f(x_0)}_{P_0(x)} + \underbrace{f'(\xi(x))(x-x_0)}_{R_0(x)}$$

Rearrange terms:

$$f'(\xi(x)) = \frac{f(x)-f(x_0)}{x-x_0}$$

- Taylor's theorem is very useful for error estimates.

Main Difficulty: Finding bounds on $R_n(x)$
→ want simple formulae for error estimates
→ simplifying often weakens the error bound

⇒ Error estimates are a balancing act between these two conflicting goals (simplicity vs strength of error bound)

Example: $f(x) = e^x$. $x_0 = 0$. Find n such that $|R_n(x)| \leq 10^{-6}$

for $x \in [0, \frac{1}{2}]$

Solution: $f^{(k)}(x) = e^x$ and $f^{(k)}(0) = 1$

$$P_n(x) = \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(0)\, x^k = \sum_{k=0}^{n} \frac{x^k}{k!}$$

$$R_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi(x))\, x^{n+1} = \frac{e^{\xi(x)}}{(n+1)!} x^{n+1}$$

Since e^x is increasing, we have

$$1 = e^0 \le e^{\xi(x)} \le e^x \le e^{1/2}$$

$$\Rightarrow |R_n(x)| \le \frac{e^x x^{n+1}}{(n+1)!} \le \frac{e^{1/2} x^{n+1}}{(n+1)!} \le \frac{e^{1/2}}{(n+1)!} \left(\frac{1}{2}\right)^{n+1}$$

- Any of these is a valid error estimate.
- The bounds get simpler but weaker as you move to the right.
- Right most is $\|\cdot\|_\infty$ maximum norm on $[0, \frac{1}{2}]$

$$\sup_{x \in [0, \frac{1}{2}]} |R_n(x)| =: \|R_n\|_\infty$$

Make a table:

n	max error
5	$3.578 \cdot 10^{-5}$
6	$2.56 \cdot 10^{-6}$
7	$1.597 \cdot 10^{-7}$

$\Rightarrow$ we need 7 terms to achieve $|R_n(x)| \le 10^{-6}$

Problem: To get an error bound, we needed to know $e^{1/2}$

Circular reasoning!

Solution: Use a different method to bound $e^{1/2}$

e.g. $e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \dots$

$$\leq 1 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = \sum_{n=0}^{\infty} \frac{1}{2^n}$$

$$= 1 + \frac{1}{1-\frac{1}{2}} = 3 \leq 4$$

$$\sqrt{e} = e^{1/2} \leq 2$$

We again simplified the error bound but made it a bit worse.